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## Relevant Information for SARAF Detector GEANT4 Simulation

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 2 attachments (7 MB)

NG\_paper\_first\_skeleton.pdf; adndt11\_1973\_\_569-619\_liskien\_t\_p\_n\_d\_d\_n\_t\_d\_n\_.pdf;

Hi,

Regarding the detector GEANT4 simulation in SARAF:

The first goal is to understand the absolute flux ratio results reported by Arik for detectors 612 and 613 during the last run, where the source acceleration voltage was 40 kV. After that, the idea is to find the proper scaling for ND3 and ND5 by estimating how many neutrons they are expected to detect and comparing that to the measured data.

I list again below the relevant information and sources:

Arik results for Run 3:

- Detector 613: distance = 99 cm, angle = 57 degrees  
 $\backslash(N_n = 7.23 \pm 0.36) \text{ neutrons/cm}^2/s$
- Detector 612: distance = 107 cm, angle = 13 degrees  
 $\backslash(N_n = 14.21 \pm 0.71) \text{ neutrons/cm}^2/s$

Arik detector sizes:

- Probably cylindrical detectors with 1 inch diameter and 2 inch length
- NE213 scintillation material
- I asked Arik for confirmation in the email you are CC'd on

Setup geometry:

- See attached picture (**theta 2 is 13 pm 1** and not 57 as in the picture)
- All detector front faces are oriented such that their normals point toward the source
- The distance from the source to the back wall (behind ND5 and detector 612) is 200 cm

ND detector properties:

- ND3 and ND5 are cylindrical detectors with:
  - 3 inch diameter
  - 3 inch length
  - EJ309 scintillation material

The current neutron rates I measure for this run are:

- ND3:  $\backslash(280.6 \pm 0.7) \text{ n/s}$
- ND5:  $\backslash(263.3 \pm 0.7) \text{ n/s}$

Please note:

- These numbers are after PSD selection, which introduces some efficiency loss (see Figure 4 in the paper draft, especially at low energies)
- Energies there are presented in quenched keV $\backslash(\text{ee})$  scale
- You can convert to proton recoil MeV scale using the mock code below (note that the input keV $\backslash(\text{ee})$  energy should be given in MeV units)

Also note:

- ND5 shows an excess of ~8% of events at higher energies than what I would naively expect from a single neutron-proton recoil interaction

- These energies are above both:
  - the maximal energy transfer expected from emitted DD fusion neutrons
  - the threshold I usually apply
- This 8% excess is *not* included in the ND5 rate written above
- I do not observe a similar excess for ND3
- I suspect this effect may be related to neutron interactions with the back wall
- Therefore, a similar phenomenon could potentially affect detector 612 as well

If useful afterward, you can also compare ND3 to detector 613 for the 50 kV acceleration voltage configuration. These two detectors were approximately at the same position during Run 2. The rate I observed there for ND3 was:

- ND3:  $(517 \pm 1)$  n/s

Arik reported:

- 613:  $(11.5 \pm 0.6)$  n/s

The paper sketch is attached, and also available here:

<https://www.overleaf.com/read/fbrffgxrnjqb#8a18d1>

I am also attaching an older paper containing:

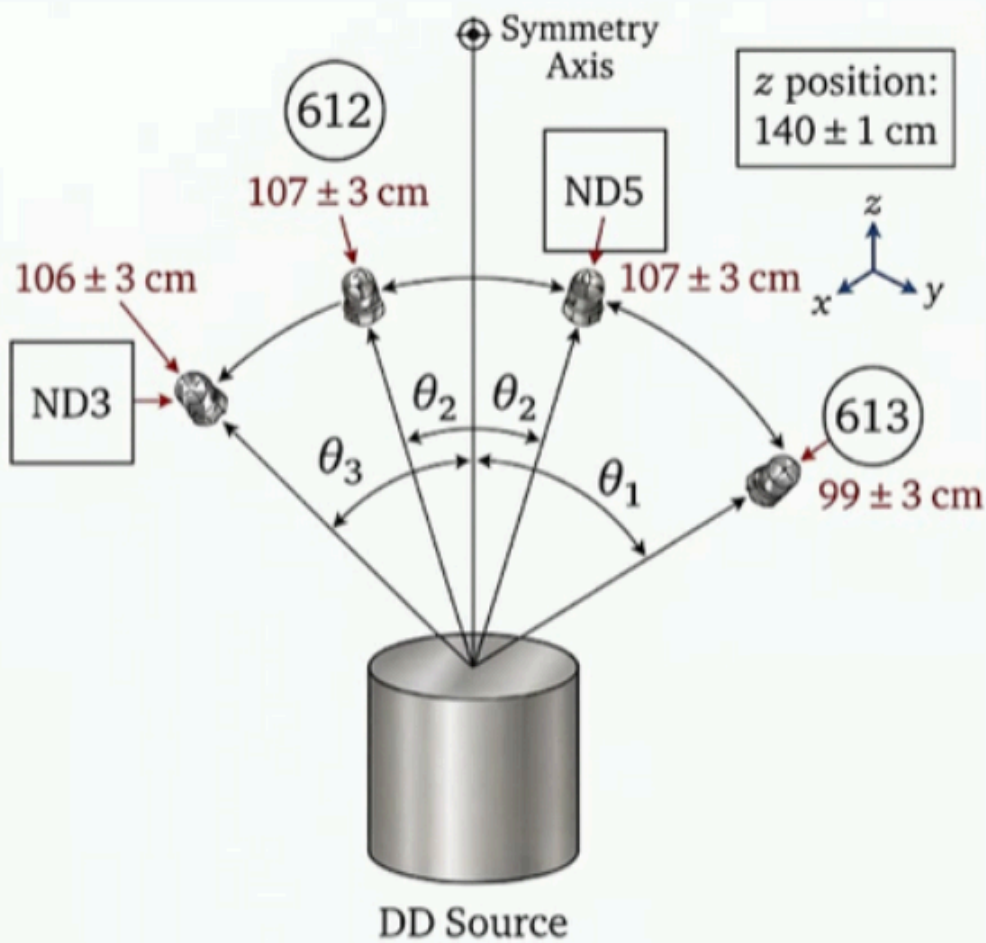
- the angular distribution dependence (Legendre polynomial expansion) for DD fusion as a function of acceleration voltage
- the neutron energy dependence on acceleration voltage

Relevant pages are 587 and 593.

Please let me know if you have any additional questions.

Thanks,

Roi



$$\theta_1 = 57 \pm 1^\circ$$

$$\theta_2 = 57 \pm 2^\circ$$

$$\theta_3 = 59 \pm 2^\circ$$

Detector Key

612, 613 ○ frame represents 1 inch detector  
 ND3, ND5 □ frame represents 3 inch detector

```
def NR_ee(E):
    A = 0.668
    B = 1.63
    C = 0.387
    D = 1.050
    return(A * E - B * (1 - np.exp(-C * (E ** D))))

from scipy.optimize import root_scalar

def find_energies_from_targets(targets):
    targets = np.asarray(targets, dtype=float)
    results = np.full(targets.shape, np.nan, dtype=float)

    for i, target in enumerate(targets):
        if not np.isfinite(target):
```

```
continue
```

```
lo, hi = 0.0, 1.0
```

```
f_lo = NR_ee(lo) - target
```

```
f_hi = NR_ee(hi) - target
```

```
while f_lo * f_hi > 0 and hi < 1e6:
```

```
    hi *= 2
```

```
    f_hi = NR_ee(hi) - target
```

```
    If f_lo * f_hi > 0:
```

```
        continue
```

```
sol = root_scalar(lambda E: NR_ee(E) - target, bracket=[lo, hi], method='brentq')
```

```
if sol.converged:
```

```
    results[i] = sol.root
```

```
return results
```

```
energies_1 = find_energies_from_targets(targets)
```

```
d_energies_1 = np.array([find_energies_from_targets(targets - d_targets[0]), find_energies_from_targets(targets + d_targets[1])])
```